

Federated Multi-Agent Deep Reinforcement Learning for Resource Allocation of Vehicle-to-Vehicle Communications

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I. INTRODUCTION

Abstract—Dynamic topology, fast-changing channels and the time sensitivity of safety-related services present challenges to the status quo of resource allocation for cellular-underlying vehicle-to-vehicle (V2V) communications. In this paper, we investigate a novel federated multi-agent deep reinforcement learning (FedMARL) approach for the decentralized joint optimization of channel selection and power control for V2V communication. The approach takes advantage of both deep reinforcement learning (DRL) and federated learning (FL), satisfying the reliability and delay requirements of V2V communication while maximizing the transmit rates of cellular links. Specifically, we elaborately construct individual V2V agent implement by the dueling double deep Q-network (D3QN), and design the reward function to train V2V agents collaboratively. As a result, each agent individually optimizes channel selection and power level based on its local observations, including the instantaneous channel state information (CSI) of corresponding V2V link, the instantaneous co-channel interference from the cellular link, the previous channels selections of nearby V2V pairs, and the queue backlog at the V2V transmitter. Another important aspect is that we incorporate FL to alleviate the training instability problem induced by cooperative multi-agent environment. The local DRL models of different V2V agents are federated periodically, addressing the limitations of partial observability on the entire network status for individual agent, and accelerating the training process of multi-agent learning. Validated via simulations, the proposed FedMARL scheme shows superiority to the baselines in terms of the cellular sum-rate and the V2V packet delivery rate.

Index Terms—V2V communication, resource allocation, deep reinforcement learning, dueling double deep Q-network (D3QN), federated learning.

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NEXT generation vehicle-to-everything (V2X) technologies have attracted considerable attention for enabling emerging vehicle-related services, which are expected to reduce road accident, enhance traffic management, and improve travel experience. Relying on various on-board sensors, critical information of surrounding environments of the vehicles can be sensed, transmitted and integrated with other vehicles in close proximity. As an important communication mode in vehicular network, vehicle-to-vehicle (V2V) communication provides the cornerstone for vehicular information exchanges between vehicles, which generally require moderate throughput but highlights stringent requirements of both delay and reliability [1]. Moreover, vehicular network significantly differ from conventional cellular network in terms of strong dynamics in channel conditions and network topologies resulting from the high mobility of vehicles [2]. Those characteristics make the design of efficient resource allocation techniques becomes challenging for V2V communication system.

Up to now, there are two prominent standards to support V2V communication, dedicated short-range communications (DSRC) [3] and Cellular-V2X (C-V2X) [4]. As the extension of conventional cellular network to the vehicular environment, C-V2X outperforms DSRC in terms of larger coverage, higher communication capability, and stronger resilience to interference [5]. In this regard, C-V2X has drawn significant attention from both academia and industry [6]–[8].

To ensure direct connectivity between neighboring vehicles, an enhanced device-to-device (D2D) interface was introduced in C-V2X systems [9], instead of forwarding vehicular packets through a base station (BS). In a D2D-underlying vehicular network, the vehicles periodically disseminate safety-critical packets while the cellular users perform high throughput communication with BS. Specially, the spectrum allocated to the cellular users could be concurrently reused by the vehicles, hence providing coexistence of different end-devices with heterogeneous quality-of-service (QoS) requirements and achieving effective utilization of limited spectrum resource [10]–[12]. However, the mutual interference generated by spectrum reuse would degrade network performance, i.e., the throughput of cellular links would decrease and the delivery of vehicular packets would fail. Besides, existing resource allocation schemes for conventional

cellular D2D communication systems, e.g., [13]–[15], cannot be directly applied to vehicular network, since those methods primarily rely on accurate channel state information (CSI) about all wireless links. However, tracing the instantaneous CSI variation of vehicular links is infeasible because of the Doppler effect resulting from the vehicles' mobility, and also imposes high signalling overhead [16]. Therefore, it is necessary to develop efficient resource allocation schemes for V2V communication to alleviate mutual interference and support heterogeneous wireless services simultaneously.

A. Related Work

Many existing studies have considered centralized resource allocation for V2V communication. The authors of [17] considered the channel uncertainty resulting from delayed CSI feedback to the BS, and analyzed the correlation of fast-changing channels. A joint channel allocation and power control algorithm was proposed to maximize system throughput while satisfying the delay and reliability requirements of each V2V link. In [18], a twin-timescale resource allocation scheme was developed, which minimizes the worst-case delay of V2V transmissions based on large-timescale road traffic information. The transmit power of each V2V link was then optimized according to small-timescale CSI. In [19], a centralized power control algorithm was designed by implement the spatio-temporal traffic pattern, which not only satisfies the delay and reliability requirements of V2V services, but also reduces the overhead of periodic CSI reports to the BS. Through the use of Lyapunov optimization, an energy-efficient resource allocation algorithm was designed in [20], subject to the queuing delay and reliability constraints for every vehicle. Graph theory has also been employed in [21]–[23] to improve system throughput. All of the above existing algorithms need the V2V links to report their local information, and would suffer from substantial signaling overhead with a growing number of vehicles. On the other hand, resource allocation has also been modeled as combinatorial optimization problems with nonlinear constraints. The problems are difficult to solve using traditional optimization methods.

Several decentralized schemes have been developed to alleviate the uncertainty of V2V communication. The authors of [24] took the density and physical proximity of vehicles into consideration, and proposed a decentralized algorithm to minimize a weighted sum of the transmission delay and improve the successful transmission probability. A three-step semi-decentralized transmission scheme was developed in [25] to mitigate packet collision and mutual interference, which consists of decentralized clustering, autonomous inter-cluster resource selection and centralized intra-cluster communication. In order to reduce transmission failures, the authors of [26] designed a decentralized retransmission mechanism to satisfy the QoS of vehicles with different priority packets. In [27], the interactions among V2V-pairs were modeled as a non-cooperative game. An oblivious equilibrium solution regulated by a sealed second-price auction was derived, and then validated both analytically and numerically. Due to the geographical separation of V2V links and the lack of the global network state, the resource

allocation decision of individual V2V link was made based on local knowledge and could hardly achieve the global optimum.

Progress has been made to integrate vehicular network and other emerging technologies, e.g., sensing, mobile edge computing (MEC), and software-defined networking (SDN). For example, the authors of [28] conceived an SDN architecture to orchestrate heterogeneous computing capabilities of vehicles, to satisfy the high reliability of delay-sensitive vehicular services. An investigation of the reliable offloading between vehicles was conducted in [29], and some typical use cases of intelligent vehicles were discussed. To cope with time-varying wireless and computation resources during vehicular movement, an adaptive offloading method was proposed to enhance network efficiency and reduce energy consumption in [30]. Moreover, multimodal sensor information can be collected and transmitted via V2V connections to improve traffic management. The authors of [31] defined a vehicular network-aided smart city architecture for the delivery of both public services and urban flow management. The techniques and application scenarios were elaborated. The authors of [32] established a V2V-aided local traffic information collection architecture, validated by a real-world trajectory dataset. A hierarchical resource management was proposed in [33] to deal with multi-source traffic data acquisition, which emphasized on the cooperation between vehicles to maximize the energy efficiency and minimize the packet drop rates.

The above existing works have emphasized on myopic optimization of resource allocation, instead of long-term performance gains. Considering the spatio-temporal variations of vehicular network, resource allocation should be modeled as sequential decision process [34]. Deep reinforcement learning (DRL) offers an effective tool to solve sequential decision problems. In [35], a resource allocation problem considering transmission mode selection of vehicular packets was studied, and a two-timescale DRL framework combined with spectral clustering was applied to obtain robust learning. In [36], a multi-agent DRL scheme was applied to allocate resources for both unicast and broadcast V2X applications in the absence of the global knowledge of the CSI. In [37], an energy-efficient resource allocation protocol was designed based on DRL to maximize the number of admissible service requests in a vehicular network, and also to conserve the battery power of roadside infrastructure. However, these studies require centralized training on data collected from all participating vehicles to obtain robust learning models. Since the data is typically decentralized among vehicles, the centralized data aggregation and training would incur significant communication overhead and raise a privacy concern.

B. Contribution and Organization

In this paper, we consider a challenging situation where the V2V links have to reuse the cellular channels already allocated to some cellular links. The V2V transmitters have to decide by themselves in a decentralized manner to choose reusable channels without interrupting cellular operations and to avoid collisions with other V2V links. We propose a federated, decentralized multi-agent deep reinforcement learning (FedMARL)

scheme, which enables the V2V pairs adjust to reusable channels and specify appropriate transmit powers. Through decentralized manner, it is avoided extra delay that would otherwise occur if traverse via the cellular infrastructure. We consider the V2V pair as an agent and elaborately construct its model with DRL, based on its instantaneous CSI and the queue backlog at the transmitter of the link, the interference from the cellular links in all channels, and the previous selections of nearby V2V links on the channels. Another important aspect is the exploitation of the federated learning (FL) technique to consolidate and synchronize the local models of the decentralized vehicular agents. By this means, each agent can acquire implicitly the knowledge of the part of the network beyond its observability to make adequate decisions on channel selection and power allocation.

The contributions of this paper are summarized as follows.

- 1) A sequential decision process problem is formulated to analyse the reliability and low delay of V2V links reusing the cellular channels while maximizing the sum-rate of incumbent cellular users. Specifically, we characterize the reliability requirement by deriving the outage probability of the V2V links in the presence of other interfering V2V and cellular links. And the Markov inequality is applied to interpret V2V transmission delay statistically and regulate the queue backlog at the vehicular transmitters.
- 2) The Dueling Double Deep Q Network (D3QN) based V2V agent is designed to enable decentralized channel selection and power allocation. The agent is tailored for practically available network knowledge, including the instantaneously measurable CSI and co-channel interference, and the outdated information of other agents' last decisions broadcast by the BS. It is capable of decreasing outage and excessive delay of the corresponding V2V links, and also preventing throughput degradation of the co-channel cellular user.
- 3) FL is adopted to synchronize the DRL models of all agents and maintain a global model, thereby addressing the partial observability of the individual agents on the network.

Extensive simulations corroborate that the new FedMARL scheme is superior to its alternatives in terms of the cellular sum-rate and the V2V packet delivery rate.

This remainder of this paper is organized as follows. Section II describes the system model and analyzes the QoS requirements of both the CUEs and V2V pairs. Section III provides the problem formulation. In Section IV, we elaborate on the proposed FedMARL scheme for the decentralized channel selection and power allocation of the V2V links. Simulation results confirm the effectiveness of the proposed algorithm in Section V, followed by conclusions in Section VI. All notations used are summarized in Table I.

II. SYSTEM MODEL

A. Scenario Description

As shown in Fig. 1, a BS serves M number of cellular users (CUEs), while there are also K pairs of V2V users (VUEs) within the coverage of the BS. The CUEs transmit with the BS via allocated cellular channels, and the transmitter of each V2V pair

TABLE I
MAIN NOTATION SUMMARY

Parameter	Value
$\mathcal{M}$	set of CUEs
$\mathcal{K}$	set of V2V pairs
W	bandwidth of each channel
t	scheduling slot index
$\zeta_{k,m}$	channel usage indicator of V2V pair k
d_0	distance threshold
φ, φ'	path loss coefficients
e	path loss exponent
$h_k^t[m]$	channel coefficient of V2V pair k over channel m in slot t
h_m^t	channel coefficient of CUE m in slot t
$h_{k,B}^t[m], h_{m,k}^t[m], h_{k',k}^t[m]$	interference channel coefficient of V2V pair k over channel m in slot t
P_m^t	transmit powers of CUE m in slot t
P_k^t	transmit powers of V2V pair k in slot t
σ^2	noise variance
$R_k^t[m]$	transmit rate of V2V pair k over channel m in slot t
R_m^t	transmit rate of CUE m in slot t
γ_0	SINR threshold of V2V links
p_o	Outage probability threshold of V2V link
λ	traffic arrival rate at V2V pair k
D_{max}	the maximal tolerable delay
Q_{max}	the maximal queue length
$\mathbb{E}[\cdot]$	expectation operator
$\Pr(\cdot)$	probability measurement

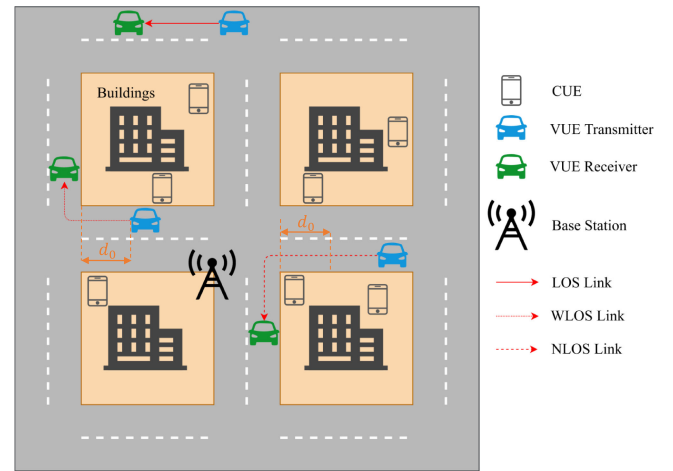


Fig. 1. An illustrative C-V2X system with single-cell setting in an urban environment, where there are multiple CUEs and V2V pairs.

sends safety-critical information to the receiver of the V2V link by reusing the allocated cellular channels. For illustration convenience, both the cellular and vehicular transceivers are equipped with a single antenna. The indexes to the CUEs and the V2V pairs are collected by $\mathcal{M} = \{1, 2, \dots, M\}$ and $\mathcal{K} = \{1, 2, \dots, K\}$, respectively.

Consider a slotted communication system. The time slots are indexed by $t = 1, 2, \dots$, and the duration of a slot is τ . The CUE links have been preassigned with orthogonal channels and fixed transmit powers, i.e., the m -th CUE occupies the m -th channel. Therefore, there are M channels with the bandwidth of each channel is W (in Hertz). The V2V pairs can reuse the channels already occupied by incumbent CUEs.

Let the binary variable $\zeta_{k,m}$ indicate the channel selection of the V2V pairs. If the k -th V2V pair selects channel m , $\zeta_{k,m} = 1$. Otherwise, $\zeta_{k,m} = 0$. We assume that the channel of each CUE can be reused by multiple V2V pairs while a V2V pair can only access to one channel at a slot, i.e.,

$$\sum_{m \in \mathcal{M}} \zeta_{k,m} \leq 1, \forall k \in \mathcal{K}; \quad (1)$$

$$\sum_{k \in \mathcal{K}} \zeta_{k,m} \leq K, \forall m \in \mathcal{M}. \quad (2)$$

B. Channel Model

At slot t , the instantaneous channel coefficient of the k -th V2V pair in channel m is

$$h_k^t[m] = \alpha_k^t \cdot g_k^t[m], \forall m \in \mathcal{M}, \quad (3)$$

where α_k^t and $g_k^t[m]$ are the large-scale fading and small-scale fading of the considered link, respectively. Due to the mobility of vehicles, the small-scale fading $g_k^t[m]$ is frequency-dependent and varies between consecutive slots, and is assumed to yield the unit-mean exponential distribution [38]. The large-scale fading α_k^t is typically frequency-independent, and changes marginally since it primarily depends on the locations of the transmitter and receiver of the V2V pair.

Depending on whether the transmitter and receiver of the k -th V2V pair are in the same lane or perpendicular lanes, the channel of the k -th V2V pair during a slot can be categorized into the following three types [39], [40]:

- 1) Line-of-sight (LOS), if both the transmitter and receiver of the V2V pair are in the same lane;
- 2) Weak-line-of-sight (WLOS), if the transmitter and receiver are in two perpendicular lanes and at least one of them is near the intersection within a predefined distance threshold d_0 ;
- 3) None-line-of-sight (NLOS), otherwise.

Let $\mathbf{x}_k = (x_{k,1}, x_{k,2}) \in \mathbb{R}^2$ and $\mathbf{y}_k = (y_{k,1}, y_{k,2}) \in \mathbb{R}^2$ denote the Euclidean coordinates of the transmitter and the receiver of the k -th V2V pair, respectively. At slot t , the large-scale fading α_k^t is given by

$$\alpha_k^t = \begin{cases} \varphi \cdot \|\mathbf{x}_k - \mathbf{y}_k\|_2^{-e}, & \text{for LOS,} \\ \varphi \cdot \|\mathbf{x}_k - \mathbf{y}_k\|_1^{-e}, & \text{for WLOS,} \\ \varphi' \cdot (|x_{k,1} - y_{k,1}| \cdot |x_{k,2} - y_{k,2}|)^{-e}, & \text{for NLOS.} \end{cases} \quad (4)$$

where $\|\cdot\|_1$ and $\|\cdot\|_2$ are the $\mathcal{L}_1$ -norm and $\mathcal{L}_2$ -norm of a vector, respectively; e is the path loss exponent; and φ and φ' are the path loss coefficients satisfying $\varphi' < \varphi(d_0/2)^e$.

The channel coefficient between the m -th CUE and BS at slot t is $h_{m,B}^t = \alpha_{m,B}^t \cdot g_{m,B}^t[m]$. Here, $\alpha_{m,B}^t$ captures the frequency-independent large-scale fading effect, including path loss and shadowing. The path loss model is defined as $128.1 + 37.6 \log_{10} d$, according to 3GPP TR 36.885 [41], where d_{cb} is the distance between the CUE and BS. The shadowing caused by obstacles affecting follows independent log-normal shadowing with standard deviation of 8 dB [41]. $g_{m,B}^t[m]$ is

the frequency-dependent small-scale fading that follows the unit-mean exponential distribution.

There are three types of interference in the considered network, as follows

- $h_{k,B}^t[m] = \alpha_{k,B}^t \cdot g_{k,B}^t[m]$: the interference channel coefficient the k -th V2V pair to the m -th CUE, when reusing the same channel m ;
- $h_{m,k}^t[m] = \alpha_{m,k}^t \cdot g_{m,k}^t[m]$: the interference channel coefficient from the m -th CUE to the k -th V2V pair, when reusing the same channel m ;
- $h_{k',k}^t[m] = \alpha_{k',k}^t \cdot g_{k',k}^t[m]$: the interference channel coefficient from the k' -th V2V pair to the k -th V2V pair ($k' \neq k$), when reusing the same channel m .

where $\alpha_{k,B}^t[m]$ and $g_{k,B}^t[m]$ denote the path loss and the small-scale fading from the k -th V2V pair to the m -th CUE, respectively, when reusing the same channel m at slot t ; $\alpha_{m,k}^t[m]$ and $g_{m,k}^t[m]$ denote the path loss and the small-scale fading from the m -th CUE to the k -th V2V pair, respectively; and $\alpha_{k',k}^t[m]$ and $g_{k',k}^t[m]$ denote the path loss and the small-scale fading from the k' -th V2V pair to the k -th V2V pair, respectively, when reusing the same channel m at slot t .

The signal-to-interference-plus-noise-ratio (SINR) of the m -th CUE can be expressed as

$$\gamma_m^t = \frac{P_m^t h_{m,B}^t}{\sigma^2 + \sum_{k \in \mathcal{K}} \zeta_{k,m} P_k^t h_{k,B}^t[m]}, \quad (5)$$

Similarly, the SINR of the k -th V2V pair when selecting channel m is given by

$$\gamma_k^t[m] = \frac{P_k^t h_k^t[m]}{\sigma^2 + I_k^{c,t}[m] + I_k^{v,t}[m]}, \quad (6)$$

where

$$I_k^{c,t}[m] = P_m^t h_{m,k}^t[m], \quad (7a)$$

$$\text{and } I_k^{v,t}[m] = \sum_{k' \neq k} \zeta_{k',m} P_{k'}^t h_{k',k}^t[m], \quad (7b)$$

are the interference from the m -th CUE and other V2V pairs reusing the same channel m , respectively; P_m^t and P_k^t are the transmit powers of the m -th CUE and the k -th V2V pair, respectively; and σ^2 is the power of the additive white Gaussian noise (AWGN) in a channel.

C. QoS Requirements of CUEs and V2V Pairs

1) *Data Rate of CUEs*: An appropriate QoS requirement of the CUEs can be parametrized as a minimum data rate to ensure smooth service experience, i.e., $R_m^t \geq R_m^{\min}$, where R_m^t is the achievable transmit rate of the m -th CUE, i.e.,

$$R_m^t = W \cdot \log_2(1 + \gamma_m^t) \quad (8)$$

Without lack of generality, we assume that each CUE has the same minimum data rate requirement.

2) *Reliability of V2V Pairs*: The reliable transmission of a V2V pair can be measured by the SINR outage probability [21]. The outage probability is the probability when the instantaneous SINR of a V2V link γ falls below a given threshold SINR

γ_0 , resulting from the fast fading of the wireless channel; i.e., $\Pr\{\gamma \leq \gamma_0\} \leq p_o$. Similar to [35] and [21], the outage probability is viewed as important metric to evaluate the reliability of V2V transmissions. The V2V link is reliable if its outage probability is lower than a certain threshold value p_o , i.e.,

$$\Pr\{\gamma_k^t[m] \leq \gamma_0\} \leq p_o \quad (9)$$

To achieve reliable transmission, the vehicles can increase their power level or switch to another channel with less interference, if their outage probability constraint cannot be satisfied. By substituting the SINR of the V2V pairs, i.e., (6), above constraint can be rewritten as

$$\begin{aligned} \Pr\{\gamma_k^t[m] \leq \gamma_0\} &= \Pr\left\{P_k^t h_k^t[m] \right. \\ &\leq \left. \sum_{k' \neq k} \zeta_{k',m}^t \gamma_0 P_{k'}^t h_{k',k}^t[m] + \gamma_0(\sigma^2 + I_k^{c,t}[m])\right\} \leq p_o \end{aligned} \quad (10)$$

Lemma 1: According to [42], if $z_1, \dots, z_n$ are independent-exponentially distributed random variables and $\mathbb{E}[z_i] = 1/\lambda_i$, $i = 1, \dots, n$, then

$$\Pr\left\{z_1 \leq \sum_{i=2}^n z_i + c\right\} = 1 - e^{-\lambda_1 c} \prod_{i=2}^n \frac{1}{1 + \frac{\lambda_1}{\lambda_i}} \quad (11)$$

where c is any positive constant.

As discussed earlier, the fast fading varies between slots and yields an i.i.d unit-mean exponential distribution. In contrast, the path loss and the transmit power remain unchanged within a slot. Therefore, $P_k^t h_k^t[m]$ and $\gamma_0 P_{k'}^t h_{k',k}^t[m]$ also yield the exponential distribution at slot t , as given by

$$\begin{aligned} \mathbb{E}[P_k^t h_k^t[m]] &= P_k^t \alpha_k^t = 1/\lambda_1; \\ \mathbb{E}[\gamma_0 P_{k'}^t h_{k',k}^t[m]] &= \gamma_0 P_{k'}^t \alpha_{k',k}^t = 1/\lambda_i, \quad i \neq 1. \end{aligned} \quad (12)$$

With Lemma 1, the outage constraint (10) is rewritten as

$$\begin{aligned} \Pr\left\{P_k^t h_k^t[m] \leq \sum_{k' \neq k} \zeta_{k',m}^t \gamma_0 P_{k'}^t h_{k',k}^t[m] + \gamma_0(\sigma^2 + I_k^{c,t}[m])\right\} \\ = 1 - \exp\left(-\frac{\gamma_0(\sigma^2 + I_k^{c,t}[m])}{P_k^t \alpha_k^t}\right) \prod_{k' \neq k} \frac{\zeta_{k',m}^t}{1 + \frac{\gamma_0 P_{k'}^t \alpha_{k',k}^t}{P_k^t \alpha_k^t}} \leq p_o \end{aligned} \quad (13)$$

By further applying the inequality in [43]: $e^k \prod_{i=1}^n (1 + z_i) \leq e^{k + \sum_{i=1}^n z_i}$, an upper bound of (13) is given by

$$\begin{aligned} 1 - \exp\left(-\frac{\gamma_0(\sigma^2 + I_k^{c,t}[m])}{P_k^t \alpha_k^t}\right) \prod_{k' \neq k} \zeta_{k',f}^t \frac{1}{1 + \frac{\gamma_0 P_{k'}^t \alpha_{k',k}^t}{P_k^t \alpha_k^t}} &\leq \\ 1 - \exp\left(-\frac{\gamma_0(\sigma^2 + I_k^{c,t}[m])}{P_k^t \alpha_k^t}\right) \prod_{k' \neq k} \zeta_{k',f}^t \left(1 + \frac{\gamma_0 P_{k'}^t \alpha_{k',k}^t}{P_k^t \alpha_k^t}\right) &\leq \\ \leq 1 - \exp\left(-\frac{\gamma_0(\sigma^2 + I_k^{c,t}[m])}{P_k^t \alpha_k^t} - \sum_{k' \neq k} \zeta_{k',f}^t \frac{\gamma_0 P_{k'}^t \alpha_{k',k}^t}{P_k^t \alpha_k^t}\right) \end{aligned}$$

$$\begin{aligned} &= 1 - \exp\left(-\frac{\gamma_0(\sigma^2 + I_k^{c,t}[m] + \sum_{k' \neq k} \zeta_{k',f}^t \gamma_0 P_{k'}^t \alpha_{k',k}^t)}{P_k^t \alpha_k^t}\right) \\ &\leq p_o. \end{aligned} \quad (14)$$

As a result, the reliability constraint of the k -th V2V pair can be written as

$$\frac{P_k^t \alpha_k^t}{\sigma^2 + I_k^{c,t}[m] + \sum_{k' \neq k} \zeta_{k',m}^t \gamma_0 P_{k'}^t \alpha_{k',k}^t} \geq \frac{\gamma_0}{\ln(\frac{1}{1-p_o})}. \quad (15)$$

3) Delay of V2V Pairs: We assume that the transmitter of a V2V link is equipped with a finite-length buffer, and the safety-related packets are generated at a constant rate λ (bits/s). As the transmit rate of the k -th V2V link $R_k^t = W \log_2(1 + \gamma_k^t)$ can differ at different slots, there can be mismatch between the packet generation and instantaneous throughput. As a consequence, the queues build up at the transmitters of the V2V pairs, hence increasing queuing delays. At the beginning of slot t , the queue length of the k -th V2V pair, denoted by Q_k^t , is given by

$$Q_k^t = \max\{0, Q_k^{t-1} + \tau\lambda - \tau R_k^{t-1}\}, \quad (16)$$

where Q_k^{t-1} is the queue length of the transmitter at slot $(t-1)$, $\tau\lambda$ is the number of bits arrived at the queue per slot, and τR_k^{t-1} is the number of bits sent to the corresponding receiver at slot $(t-1)$.

We consider the queuing delay which dominates the delay over a V2V link [44]. According to Little's Law [20], the average queuing delay is linear to the queue length. Let $D_{\max}$ denote the tolerable transmission delay for V2V packets, and the amount of packets experiencing longer delays than $D_{\max}$ can be written as $Q_{\max} = \lambda D_{\max}$. In this sense, the delay constraint of the k -th V2V pair can be rewritten to ensure a steady-state queue length with tolerable probability threshold.

$$\Pr\{D_k^t \geq D_{\max}\} = \Pr\{Q_k^t \geq Q_{\max}\} \leq p_o \quad (17)$$

By employing the Markov's inequality, i.e., $\Pr(X \geq a) \leq \mathbb{E}[X]/a$ if X is a non-negative random variable and $a > 0$ [20], we can tighten the queue length constraint, i.e., (17), as follows.

$$\bar{Q}_k = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbb{E}[Q_k^t] \leq p_o \cdot Q_{\max}. \quad (18)$$

The V2V packets can be delivered in time if the time-averaged queue length of the transmitter is upper bounded.

III. PROBLEM FORMULATION

We aim to maximize the time-averaged transmit rates of the CUEs while guaranteeing the reliability and delay requirements of the V2V pairs. The maximum V2V transmission delay is T , e.g., 100 milliseconds, as defined in 3GPP TR 36.885 [41]. We focus on the decentralized channel access and power control for the V2V pairs in the situation where the BS has centrally assigned the CUEs with orthogonal OFDM subchannels. The m -th CUE has been assigned with the m -th channel. This is due to the fact that the priority is typically given to the CUE links (at channel allocation). In this sense, the CUE's channel allocation is a known parameter in the considered system model, and not

a part of the optimization variables. The considered problem is formulated as

$$\max_{\zeta_{k,m}, P_k^t, \forall k, m} \frac{1}{T} \frac{1}{M} \sum_{t=1}^T \sum_{m=1}^M W \cdot \log_2(1 + \gamma_m^t), \quad (19a)$$

$$\text{s.t. } (15) \& (18), \forall k \in \mathcal{K}, \quad (19b)$$

$$R_m^t \geq R_m^{\min}, \forall m \in \mathcal{M}, \quad (19c)$$

$$0 \leq P_k^t \leq P_{\max}^v, \forall k \in \mathcal{K}, \quad (19d)$$

$$Q_k^t \leq Q_{\max}, \forall k \in \mathcal{K}, \quad (19e)$$

$$\sum_{m \in \mathcal{M}} \zeta_{k,m}^t \leq 1, \forall k \in \mathcal{K}, \quad (19f)$$

$$\sum_{k \in \mathcal{K}} \zeta_{k,m}^t \leq K, \forall m \in \mathcal{M}, \quad (19g)$$

$$\zeta_{k,m}^t \in \{0, 1\}. \quad (19h)$$

Problem (19) is a time-averaged combinatorial optimization problem, which is mathematically challenging and computationally prohibitive. Typical centralized solutions are inadequate due to difficulties in acquiring the instantaneous CSI of all links. To address these issues, we propose to exploit DRL to solve problem (19) in a decentralized fashion. In what follows, we transform the problem of joint channel selection and power allocation into a multi-agent DRL problem. All V2V pairs act as agents, independently performing and updating their resource allocation strategy.

IV. PROPOSED DECENTRALIZED RESOURCE ALLOCATION BASED ON FEDMARL

We introduce the basic concepts of DRL and its extension to multi-agent setting. With appropriate DRL algorithms, an intelligent agent could be trained to perform optimal sequential decisions through interacting with the environment via trial and error. At time step t , the agent observes its surrounding environment and obtains observation as $s^t \in \mathcal{S}$, and then executes action $a^t \in \mathcal{A}$, according to a certain policy $\pi: \mathcal{S} \rightarrow \mathcal{A}$, which is the probability of an action a being taken under state s . The environment is influenced by the executed action and subsequently transfers to the next state $s^{t+1} \in \mathcal{S}$. As a result, a reward $r^t = \mathcal{R}(s^t, a^t)$ evaluating the effect of action a^t is provided to the agent for adjusting individual policy. This interaction with the environment at time step t forms an experience that is described by a tuple (s^t, a^t, r^t, s^{t+1}) .

A widely used DRL algorithm is Q-learning, which can maximize the expected cumulative reward (Q-value) based on policy $\pi: Q_\pi(s, a)$, i.e.,

$$Q_\pi(s, a) = \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \beta^k r^{t+k+1} \middle| s^t = s, a^t = a \right] \quad (20)$$

where $\beta \in [0, 1]$ is a discount factor reflecting the impact of the future reward, and the expectation operation $\mathbb{E}_\pi[\cdot]$ is taken over

all possible state-action transitions following the policy π . Q-learning requires the agent to maintain a lookup table (i.e., the Q-table) to store the optimal Q-value $Q^*(s, a) = \max_\pi Q_\pi(s, a)$ for all possible state-action pairs. The Q-value is updated iteratively. At the i -th exploration step, the update process follows

$$\begin{aligned} \delta^{(i)} &= r + \beta \max_{a'} Q^{(i)}(s', a') - Q^{(i)}(s, a); \\ Q^{(i+1)}(s, a) &\leftarrow Q^{(i)}(s, a) + \varepsilon \cdot \delta^{(i)}, \end{aligned} \quad (21)$$

where s' and a' are the state and action at the next time step, respectively. δ is the temporal-difference (TD) error. The learning rate ε corrects the TD error such that the Q-table can approximate $Q^*(s, a)$. However, Q-learning can be hardly applicable to large-scale problems, because Q-learning is expensive in terms of computational complexity and storage to update and store Q-tables with high-dimensional state and action spaces. DRL can overcome this issue by adopting a neural network $Q(s, a; \theta_e)$ to approximate the optimal Q-value, where θ_e denotes the parameters of the neural network. The mapping from a state to an action is done by leveraging a non-linear approximation of the neural network, and the update process of the original Q-table is transformed to the update of the neural network parameters.

In the presence of multiple agents, the environment is influenced by the joint actions of the agents. Each agent executes its individual action a_k^t based on its local observation of the whole environment. The joint action $\mathcal{A}^t = (a_1^t, a_2^t, \dots, a_K^t)$ collectively influences the common environment and determines the next state $\mathcal{S}^{t+1}$. In this paper, different agents compete for the channels, and behave selfishly in pursuit of their distinct reward maximization, rather than achieving global performance. For this reason, we are interested in decentralized DRL model training with a collaborative reward setting, where all agents have a common reward and the learning objective is to jointly maximize the global cumulative reward of all agents.

A. Agent Design

The key elements of the proposed learning model are described in detail, as follows.

1) *State*: The global state of the overall vehicular network $\mathcal{S}^t$ consists of the channel conditions of all CUEs and V2V pairs, and the resource allocation actions of the V2V pairs. The global state is unknown to the individual agents. It was reported in [35] and [36] that integrating the information about the channel utilization of geographically adjacent V2V pairs can benefit resource allocation, since those vehicles are likely to have similar channel states due to their geographical proximity. In light of this, we define the neighborhood of the k -th V2V pair as the area within the maximum communication radius of the k -th V2V pair. (The maximum communication radius is set to 300 m in our simulations, according to 3GPP TR 36.885 [41].) The state space of any V2V agent k is defined to contain:

- The channel coefficient of the V2V pair, $h_k^t[m]$, $m \in \mathcal{M}$;
- The channel coefficients of the CUEs, $h_{m,B}^t[m]$, $m \in \mathcal{M}$;
- The channel selection of the neighbors of the k -th V2V pair at the last slot, i.e., $[N_k^{1,t-1}, \dots, N_k^{M,t-1}]$, where $N_k^{m,t-1}$

denotes the number of neighboring V2V pairs that reused channel m in slot $(t - 1)$;

- The current queue length at the transmitter of the k -th V2V pair, i.e., Q_k^t .

The scale of the state space is $3M + 1$ per V2V pair (or agent), depending on the number of available channels in the system. The V2V agents can acquire the state information with the assistance of limited signaling overhead. The channel coefficient $h_k^t[m]$ is estimated by the receiver of the V2V link at the beginning of each scheduling slot, and fed back to the transmitter. The channel coefficients of the CUEs $h_{m,B}^t[m]$ are estimated by the BS and broadcast to all vehicles within its coverage [45]. The previous channel selection of the neighboring V2V pairs can be obtained via radio sensing or sidelink control channel. The queue length Q_k^t is readily available to the transmitter of the V2V pair.

2) *Action*: The action executed by agent k , i.e., the k -th V2V pair, includes the channel selection $\zeta_{k,m}$, and the transmit power P_k . The transmit power is discretized into $N_p + 1$ levels, i.e., $\{0, \frac{1}{N_p} P_{\max}^v, \frac{2}{N_p} P_{\max}^v, \dots, P_{\max}^v\}$. No packet is transmitted over a V2V link, if the transmit power is zero. As a result, the dimension of the action space is $M \times (N_p + 1)$.

3) *Reward*: To maximize the sum-rate of all CUEs and guarantee the QoS requirements of both the CUEs and the V2V pairs, we design the reward function as

$$\begin{aligned} \mathcal{R}^t = & \lambda_1 \sum_{m \in \mathcal{M}} U(R_m^t - R_m^{\min}) + \lambda_2 \sum_{k \in \mathcal{K}} U(Q_k^t - Q_{\max}) \\ & + \lambda_3 \sum_{k \in \mathcal{K}} U \\ & \times \left(\frac{P_k^t \alpha_k^t}{\sigma^2 + I_k^{c,t}[m] + \sum_{k' \neq k} \zeta_{k',f}^t P_{k'}^t \alpha_{k',k}^t} - \frac{\gamma_o}{\ln(\frac{1}{1-p_o})} \right) \end{aligned} \quad (22)$$

Here, $U(x)$ provides a penalty if any part of the reward is unsatisfied, as given by

$$U(x) = \begin{cases} x, & \text{if } x > 0; \\ A, & \text{if } x \leq 0, \end{cases} \quad (23)$$

where $A < 0$; and λ_1 , λ_2 and λ_3 are the weighting coefficients of the three parts of the defined reward. At time slot t , all agents take their actions independently, and the global reward $\mathcal{R}^t$ is evaluated at the BS. The agents obtain the evaluation of the global reward at the end of every slot, broadcast by the BS with small signaling overhead.

B. Local Training of V2V Agents

We build the DRL model for the V2V pairs by leveraging the Dueling Double Deep Q Network (D3QN) architecture [46] and priority experience replay (PER) technique [47]. D3QN is an extension of the Double Deep Q Network (DDQN) with the following adjustments to the neural network structure:

- In the DDQN model, the output layer of the neural network produces the action value function $Q(s, a_i)$ directly for all possible actions, to evaluate the gain of action a_i taken at state s .
- By contrast, the D3QN model integrates the dueling architecture to the DDQN. The output layer is split into two

subnetworks to separately estimate the state value $V(s)$ and the advantage function $A(s, a_i)$. The value of $V(s)$ is a scalar to obtain the estimation of the current state s . $A(s, a_i)$ is a vector with the dimension of the action number, and each element evaluates the advantage of action a_i over the other possible actions. In the end, the output action value $Q(s, a_i)$ is estimated by combining the state value of the current state, $V(s)$, and the advantage of each action in the state, $A(s, a_i)$.

The structure of DRL model is illustrated as Fig. 2. There are two neural networks per V2V agent, i.e., the evaluate network and the target network with parameters θ_e and θ_t , respectively. The evaluate network makes the resource allocation decision a^* based on the current state s , and the ϵ -greedy mechanism is employed, i.e., the agent chooses action $\arg \max_a Q(s, a; \theta_e)$ which owns the maximal estimated Q-value with probability $1 - \epsilon$. Otherwise, a random action is selected with probability ϵ . After the selected action is executed, the state that the agents observe turns into state s' and a reward is returned. By applying the PER technique, the experience of the agent $h = (s, a, r, s')$ is stored into the memory, and the target network calculates the expected value $Q(s', a; \theta_t)$ for the next state. The priority of an experience j , denoted by $P(j)$, is evaluated according to the TD errors δ_j :

$$P(j) = \frac{p_j^\alpha}{\sum p_j^\alpha}, \quad p_j = 1/\text{rank}(\delta_j), \quad (24)$$

where α is a hyper-parameter to influence priority. In each update phase, a mini-batch of B numbers of collected historical experiences, denoted by $\mathcal{B}$, is sampled from the replay memory, where the samples with higher priorities are drawn more frequently. It is noted that if $\alpha \rightarrow 0$, no priority is given and uniform sampling is carried out in the memory. The loss function $\mathcal{J}(\theta)$ is determined by the network parameters $\theta = \{\theta_e, \theta_t\}$. It is a weighted average over all samples:

$$\begin{aligned} \mathcal{J}(\theta) = & \frac{1}{B} \sum_{h \in \mathcal{B}} \mathcal{J}(\theta; h) \\ = & \frac{1}{B} \sum_{h \in \mathcal{B}} \left(r + \beta \max_a Q(s', a; \theta_t) - Q(s, a^*; \theta_e) \right)^2 \end{aligned} \quad (25)$$

In the designed D3QN algorithm, the two sets of parameters are not updated at the same pace. The parameters of the target network θ_t are fixed while the evaluate network θ_e is updated. The parameters are only updated after the evaluate network is updated C times ($C = 4$ in simulation setting). A soft update factor $\tau \leq 1$ is used, i.e., $\theta_t \leftarrow \tau \theta_e + (1 - \tau) \theta_t$, thus stabilizing the learning process. Algorithm 1 summarizes the training procedure.

C. Federated Learning Design for Model Training

The workflow of the FL for solving the V2V resource allocation problem is shown in Fig. 2. In the considered vehicular network, the BS maintains a global DRL model, and all V2V agents constitute their local models with the same network structure.

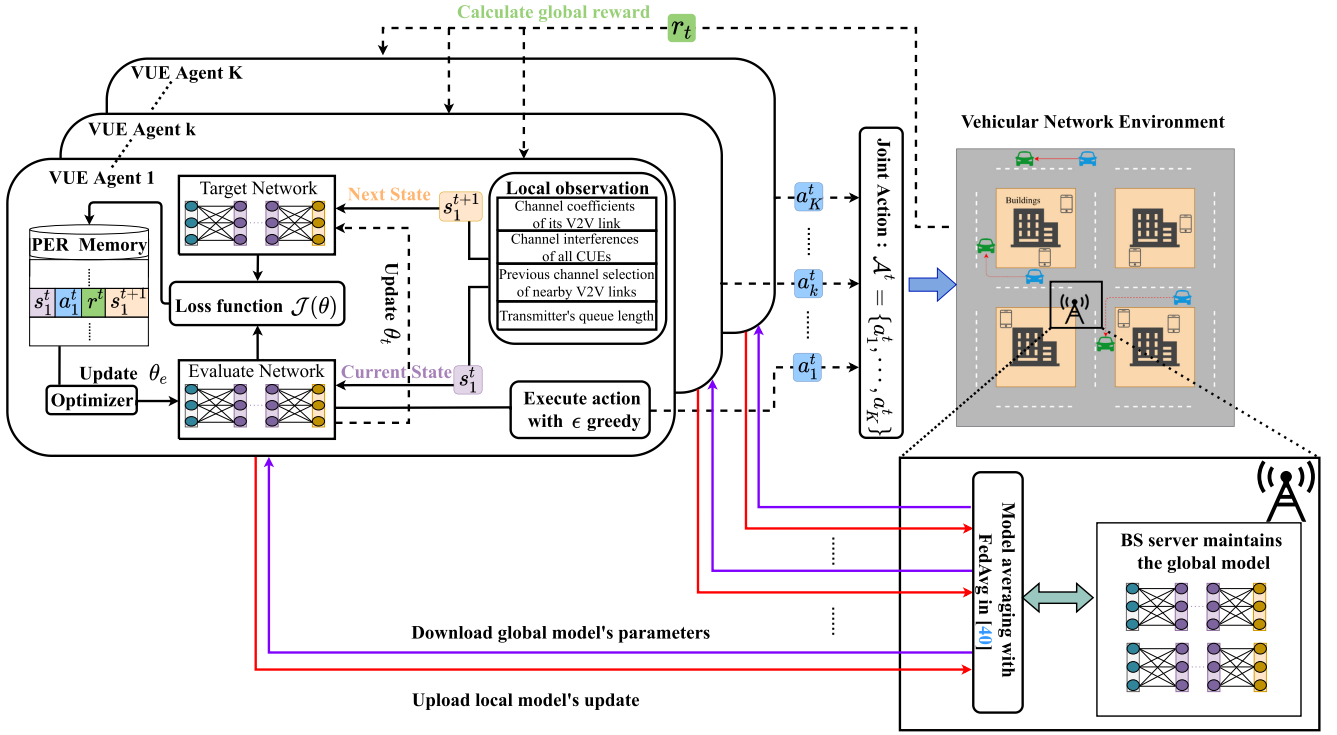


Fig. 2. The proposed FedMADRL architecture for V2V resource allocation.

We adopt a typical FL algorithm, named FedAvg [48], for the model iteration. FedAvg divides the training process of a global model into rounds. In each round indexed by g , a local model is updated with a random mini-batch of data samples at each V2V agent in the way as described in Section IV-B. The local update of the k -th V2V agent minimizes the aforementioned loss function, $\mathcal{J}_k(\theta)$, with its mini-batch sample B_k . At the end of a round, the global model is obtained by taking a weighted average of the parameters of all the local models involved in the learning. The minimization of the global loss function $\mathcal{J}_k(\theta)$ follows

$$\min_{\theta \in \mathbb{R}} \mathcal{J}(\theta) = \sum_{k=1}^K p_k \mathcal{J}_k(\theta) \quad (26)$$

where $p_k = \frac{|B_k|}{\sum |B_k|}$ is the weight of the k -th V2V agent. Since the states observed by each V2V link cannot fully characterize the entire vehicular environment, FL provides an effective method for improving the model performance with decentralized local DRL models [49]. In any iteration, i.e., the g -th iteration, the V2V agents interact with the BS which serves as the model aggregator, as follows.

1) *Local Update*: To obtain its local model, each V2V agent first receives the latest parameters of the global model $\theta^{(g-1)} = \{\theta_e^{(g-1)}, \theta_t^{(g-1)}\}$ from the BS server. Then, the V2V agent computes its gradient on the historical experience and updates its local model by

$$\theta_k^{(g)} = \theta^{(g-1)} - \eta \cdot \nabla \mathcal{J}_k(\theta)^{(g)} \quad (27)$$

where η is the learning rate. To minimize the loss function, stochastic gradient descent (SGD) is used to provide a stochastic approximation of the gradient descent.

2) *Upload*: After the local update, the k -th V2V agent sends $\theta_k^{(g)}$ to the BS.

3) *Aggregation and Feedback*: Upon the receipt of all the uploaded local models, the BS aggregates the models (i.e., taking a weighted average of the local models) and obtains the updated global model, as follows:

$$\theta^{(g)} = \sum_{k=1}^K p_k \theta_k^{(g)} \quad (28)$$

The BS broadcasts the global model parameters $\theta^{(g)}$ to all V2V agents.

These three steps repeat until the required accuracy of the learning model is achieved. According to [50] and [51], for a given learning accuracy $0 < \varsigma < 1$, after $\mathcal{O}(\log(\frac{1}{\varsigma}))$ rounds, the FedAvg algorithm can achieve a global model convergence when $|\mathcal{J}(\theta^{(g)}) - \mathcal{J}(\theta^*)| \leq \varsigma, \forall g \geq \log(\frac{1}{\varsigma})$. Here, θ^* is the optimal solution for minimizing the global loss function. The computational complexity of FedAvg accounts for the local update of the network parameters via SGD, and the model aggregation at the BS. Since SGD approximates the gradient with a single example and sweeps through all samples, the complexity of the local update is $\mathcal{O}(BK)$, depending on the numbers of mini-batch samples B and the number of participating agents K . The complexity of the model aggregation is $\mathcal{O}(K)$, as it grows linearly with the number of agents K . Consider that there are $\mathcal{O}(\log(\frac{1}{\varsigma}))$ rounds until the convergence of the global model. The overall complexity of the FedAvg algorithm is $\mathcal{O}((BK + K) \cdot \log(\frac{1}{\varsigma}))$.

Algorithm 1: Training Procedure for Each V2V Agent.

```

1: Randomly initialize the evaluate network with
   parameters  $\theta_e$ , and set parameters  $\theta_t = \theta_e$  for the
   target network.
2: Initialize the prioritized replay memory.
3: for each episode  $epi = 1, 2, \dots$  do
4:   Initialize communication links for CUEs and V2V
   pairs, and get initial environment state.
5:   for each time slot  $t = 1, 2, \dots$  do
6:     All agents select action  $a^t$  according to  $\epsilon$  greedy
     policy based on observation of current
     environment state  $s^t$ , and choose action  $a^t$ ;
7:   end for
8:   All agents take actions, and the BS calculates global
   reward  $r^t$  as (22) according to the joint action;
9:   for each agent do
10:    Get next observation  $s^{t+1}$ ;
11:    Store the experience  $(s^t, a^t, r^t, s^{t+1})$  into replay
    memory for each agent;
12:   end for
13:   if over every  $f_e$  slots then
14:     for each agent do
15:       Sample a mini-batch of historical experiences
       and update  $\theta_e$  by optimizing loss function (25)
       with stochastic gradient descent (SGD)
16:     end for
17:   end if
18:   if over every  $f_t$  slots then
19:     for each agent
20:       Update target network parameters  $\theta_t$ 
21:     end for
22:   end if
23: end for

```

V. EXPERIMENTAL RESULTS

In this section, we present numerical experiments to evaluate the proposed algorithm. The Pytorch-based simulations are implemented on a PC with a CPU (Intel(R) Xeon(R) Gold 6134 M) and a GPU (NVIDIA Quadro P5000). The memory of the PC is 32 GB. We adopt the urban scenario specified in 3GPP TR 36.885 [41], where the BS is located at the center of the network. There are 15 CUEs preassigned dedicated channels for transmissions to the BS. The vehicles are dropped randomly in lanes, traveling at a fixed speed along the lines. Each vehicle disseminates safety-critical packets periodically, and the maximum tolerable transmission delay of a V2V packet is set to 100 ms. The other simulation parameters are summarized in Tables II and III.

To verify the efficiency of the proposed algorithm, the following methods are simulated for performance comparison.

- 1) Greedy channel allocation and power control method (GM): In this method, the V2V agents always select the channel with the lowest interference for V2V transmission. The transmit power is set to the maximum level.

TABLE II
SIMULATION ENVIRONMENT PARAMETERS

Parameter	Value
Carrier frequency	2 GHz
Number of channels	15
Bandwidth of each channel	1 MHz
Number of CUE	15
Number of V2V pairs	[5,10,15,20,25,30]
Antenna height of BS	25 m
BS antenna gain	8 dBi
Antenna height of CUE/VUE	1.5 m
Vehicle antenna gain	3 dBi
SINR threshold for VUEs γ_o	5 dB
Transmit power of CUE	23 dBm
Transmit power of VUEs	$[-\infty, 5, 20, 23]$ dBm
Noise power σ^2	-174 dBm/Hz
V2V packet size	$[1, 2 \dots] \times 1060$ bytes
Path loss coefficients φ'	-54.5 dB
Path loss coefficients φ	-68.5 dB
Distance threshold d_0	15 m
Path loss exponent e	1.61

TABLE III
DRL MODEL PARAMETERS

Parameter	Value
Replay memory size	10,000
Mini-batch size d_{cb}	128
Start exploration rate	1
Final exploration rate	0.1
Optimizer	Adam
Learning rate η	0.7
Active function	ReLU: $f(x) = \max(0, x)$
Number of neurons per layer	[500, 250, 120]
Weights of reward function	0.3, 0.5, 0.2
Frequency of update evaluate network	100 slots
Frequency of update target network	400 slots

- 2) Cluster-based resource allocation method (CM) [20]: The vehicles are first grouped based on their geographical proximity. The channels are allocated separately in each group, and can be reused between different groups. The vehicles optimize their transmit powers locally to minimize the intra-group interference. The predefined number of clusters is set to 5.
- 3) Single-agent DRL (SADRL): In this method, only a single V2V pair updates its DRL model for resource allocation based on its locally acquired information. The other V2V agents replicate the DRL model with the coordination of the BS.

A. Complexity Analysis

In this section, we discuss the complexity of our proposed algorithm with the baseline methods.

- 1) GM: The computational complexity of this method is dominated by the enumeration of all M available channels for the one with the lowest interference. The fast-sorting algorithm is implied at the BS to compare all M channels, incurring the complexity of $\mathcal{O}(M \log(M))$. Therefore, the total computational complexity of all V2V pairs is $\mathcal{O}(K \cdot M \log(M))$.
- 2) CM: This scheme first clusters the V2V pairs into disjoint zones based on the locations of the vehicles. Then, the resource allocation is optimized in each cluster. The

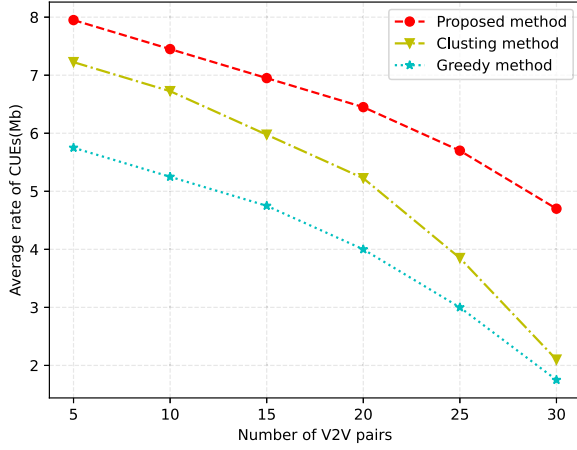


Fig. 3. Average rate of the CUEs vs. different number of V2V pairs.

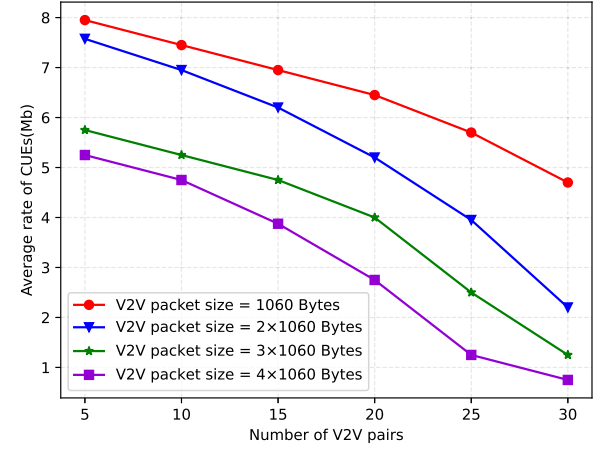


Fig. 4. Average rate of the CUEs vs. different number of V2V pairs, under different V2V packet sizes.

complexity is made up of two parts. The first part accounts for the clustering process. As considered in [20], a density-based spatial clustering algorithm [52] is applied. Its complexity depends on the number of clusters. Let C denote the number of clusters, the computational complexity of the clustering process is $O(C^2)$. The second part of the complexity accounts for the resource allocation in each cluster. A so-called deferred acceptance (DA) algorithm [53] is adopted to find the stable matching between wireless resources and V2V pairs in a cluster. It incurs the computational cost of $O(M_c \cdot K_c)$ in the c -th cluster, where M_c and K_c are the number of channels and V2V pairs in the c -th cluster, respectively. Therefore, the total complexity of the CM is $O(C^2 + \sum_{c=1}^C M_c \cdot K_c)$.

3) SADRL and proposed FedMARL: The DRL-based method is dominated by the update process of the network parameters. Suppose that the DRL agent consists of H hidden layers with x_h neurons in the h -th layer. Let Z be the dimension of the input layer, i.e., the dimension of the state space, which is proportional to the number of V2V pairs K . The output layer of the DRL model provides the resource allocation actions, including the selection of $N_p + 1$ transmit power levels and M available channels. Under the double neural network setting, the total number of model parameters is: $X = 2(Zx_1 + \sum_{h=1}^H x_h x_{h+1} + M(N_p + 1))$. The update process of the loss function is linear to the mini-batch size B since the SGD optimizer scans all mini-batch samples. As a result, an update of all parameters incurs a computational complexity of $O(BX)$. In every model update phase, the computational complexity of the SADRL method is $O(BX)$, since only a single V2V agent updates its model. Consider the proposed FedMARL method, the computational complexity is $O(KBX)$ for updating the network parameters of all K agents.

B. Average Rate Performance of CUEs

Fig. 3 shows that the average rate of the CUEs declines with the increasing number of V2V pairs, since more V2V pairs have

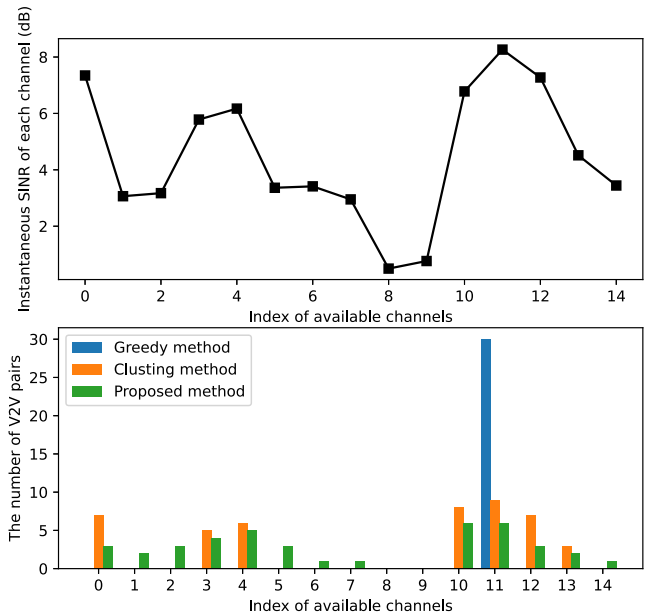
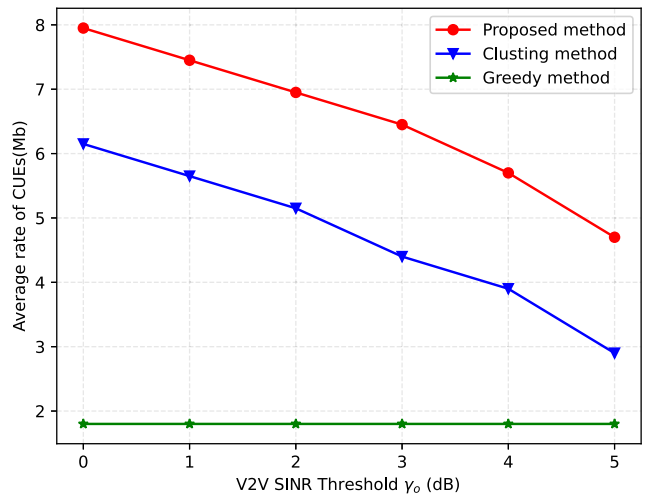


Fig. 5. The proportion of V2V agents' selection on available channels.

Fig. 6. Average rate of CUEs versus V2V SINR threshold γ_o .

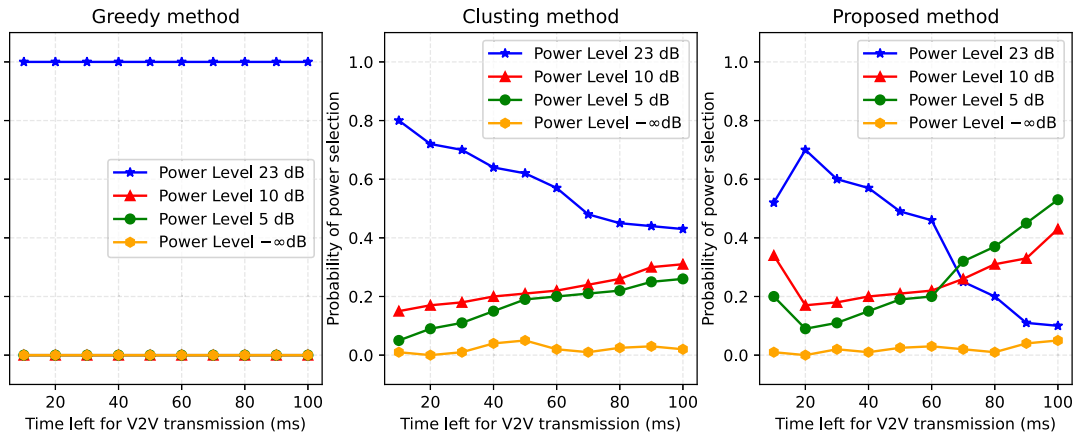


Fig. 7. The distribution of power-level with the remaining transmission time.

to share the same channel. As a result, the interference from those V2V links becomes severer, and the CUE links suffer from losses of the transmit rates. Nevertheless, the proposed method can better mitigate the interference by coordinating the vehicles on different channels and choosing appropriate power levels based on local observations. The greedy method performs the worst because it requires the V2V pairs to utilize their maximum power for transmission, deteriorating the transmit rates of the CUEs. In the case of high traffic density, there is almost no data transmission for the cellular links.

The clustering method developed in [20] firstly divides all V2V pairs into different clusters and then optimizes the channel selection and transmit power for the V2V pair in each cluster. It is hard to determine the appropriate number of clusters because of the time-varying number of V2V pairs, especially in large-scale vehicular network. With the increasing number of V2V pairs, the clustering method [20] slightly outperforms the greedy method but is still worse than the proposed method. The performance gap between the proposed and the clustering methods increases, confirming the robustness of the proposed algorithm to the time-varying traffic density.

Fig. 4 shows that our algorithm achieves highly desirable CUE performance under different V2V packet sizes. It can be seen that under different V2V packet sizes, the average rate of the CUE links decreases slowly when the number of V2V pairs is smaller than the number of CUE links. The reason is that there are plenty of unselected channels that can be allocated, and the CUE links undergo mild interference. The average rate of the CUE links changes slowly with the increasing packet sizes of the V2V links. This is due to the fact that high transmit power levels have to be selected to guarantee the QoS of the V2V links.

C. Analysis of Chanel Selection

Fig. 5 shows the distribution of the channel selection made by the V2V agents, where there are 30 V2V agents. In the upper subfigure of Fig. 5, the instantaneous SINR is provided for each channel. The higher SINR value means the corresponding channel (i.e., channel 11) is not heavily reduced and undergoes less interference. Since the greedy method always selects the channel with the lowest interference for V2V transmission, all

V2V pairs join this channel. The proposed method outperforms the clustering method since it can reuse the channels more evenly. The channels with moderate interference (i.e., channels 1, 2, 5, 6, 7, and 14) can also be chosen by the V2V agents for transmission. In contrast, the clustering method focuses on the better channels (i.e., channels 0, 3, 4, 10–13).

D. Analysis of V2V SINR Threshold

Fig. 6 shows the average rate of the CUEs versus different SINR threshold γ_o . The number of V2V pairs is 10, and the V2V packet size is 2120 bytes. It is observed that with the increase of γ_o , the average rate of the CUEs declines. This is because a larger γ_o requires the V2V pairs to select a higher transmission power level to guarantee the reliability requirement (15), resulting in stronger interference to the CUEs sharing the same channel. It is also seen that in the greedy method, a V2V pair always selects its maximum transmission power without considering the transmit rate of the CUEs. Thus, the average rate of the CUEs remains unchanged for different outage thresholds. The proposed method outperforms the two other methods, and can effectively alleviate the interference by adaptively selecting the best reusable channels and adequate transmission powers.

E. Analysis of Transmission Power

Fig. 7 plots the probability of a V2V agent selecting different transmit power levels versus the remaining transmission time of the V2V packets. The greedy method always selects the maximum power level, that is, the probability of selecting 23 dB is 1, and the probabilities of selecting any other power levels are zero. The clustering method keeps a high probability of choosing the maximum power. However, when there is sufficient transmission time, the clustering method [20] would lead to strong interference and performance degradation. That is because the resource allocation by the clustering method only optimizes individual clusters. In contrast, our proposed method enables all V2V agents to learn the effective transmit power levels in response to the remaining transmission time of the V2V links.

When sufficient transmission time remains, the V2V agents preferably choose lower transmit power levels to reduce resource

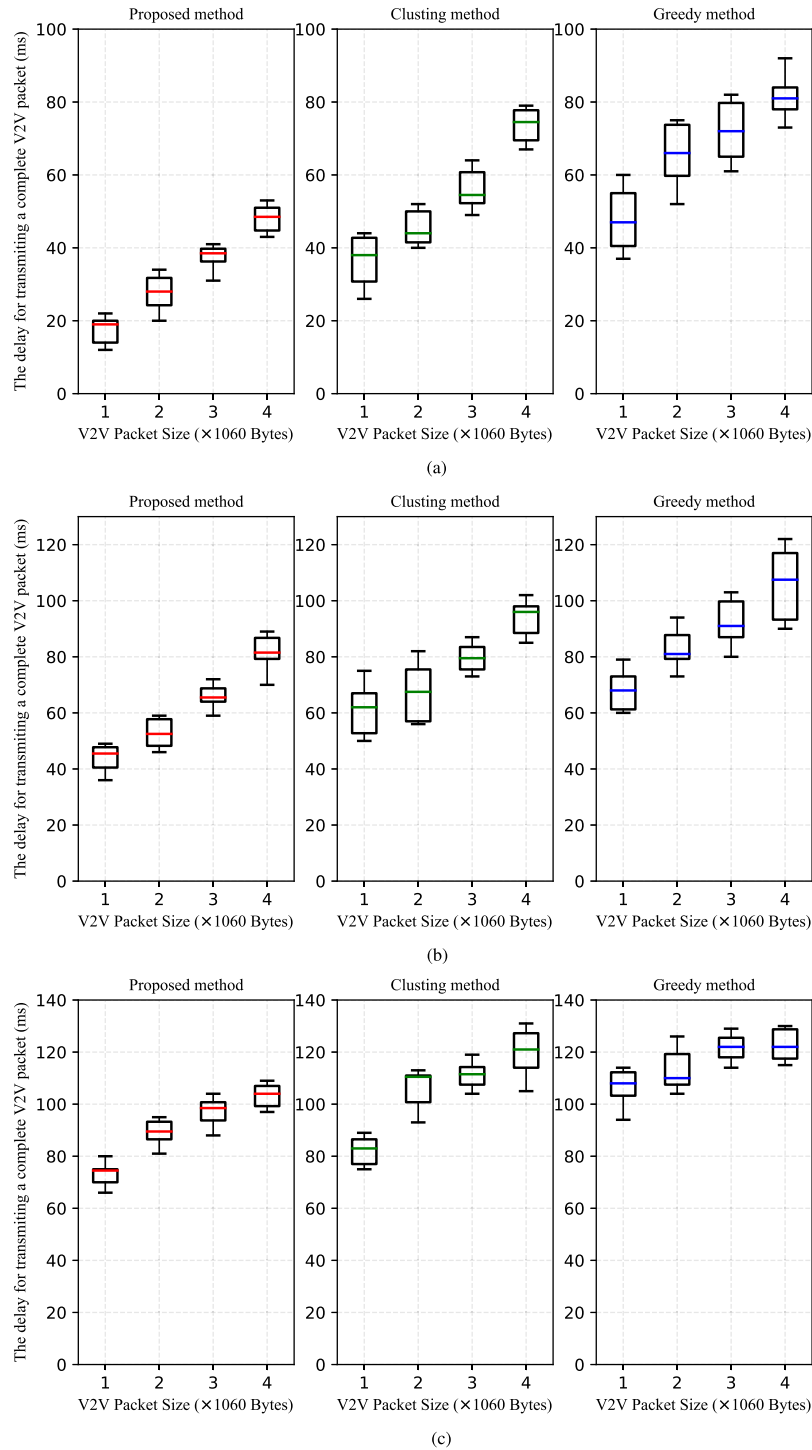


Fig. 8. The distributions of transmission delay of various V2V packets size under different methods. (a) The number of V2V pairs is 5. (b) The number of V2V pairs is 15. (c) The number of V2V pairs is 30.

consumption and interference to the other vehicles. This can be seen on the right of the cross point, when the remaining

transmission time is 70 ms. There is also a turning point on the curve that always chooses the highest transmit power level. The probability of selecting the highest transmit power level drops rapidly, when there is only 10 ms left for a V2V packet. This is because the agent learns that even with the maximum transmit power, it still cannot meet the requirements of transmission

delay. The selection of a lower transmit power level can offer a higher reward, due to the reduced interference to the CUE and other V2V links.

F. Analysis of V2V Transmission Delay

The boxplots shown in Fig. 8 depict the distribution of packets transmission delay among all V2V pairs with respect to the

increasing V2V packet size. The lower border of a boxplot is the minimum among all V2V pairs, and the upper border is the maximum. Besides, the colored horizontal line drawn in the middle gives the average delay. We compare our proposed method with the clustering method [20] and the greedy method under different numbers of V2V pairs. In our simulation, the number of CUEs is set to 15 (i.e., the number of available channels). The number of V2V pairs is set to 5, 15, and 30, representing different degrees of traffic density and competition for available channels.

Fig. 8 shows that, given a fixed number of V2V pairs, the transmission delays of all methods increase with the packet size, because a larger size of the V2V packets results in a longer V2V transmission duration and a higher V2V transmit power. Several observations are made in Fig. 8 to confirm the merits of our proposed method: 1) the average transmission delay (the colored horizontal line) of the proposed method is shorter than those of the other two methods; and 2) the proposed method undergoes smaller fluctuations in the transmission delay of all V2V pairs, i.e., the vertically smaller gap of the boxplots. To this end, the proposed algorithm is robust to the varying packet sizes.

It is also observed that even under the same V2V packet sizes, the increasing number of V2V pairs leads to a longer transmission delay, due to severer interference. The transmit rates of the V2V pairs decline. There is a slight increase of the delay under the proposed method, as compared with the other two methods. The reason is that the multi-agent architecture of the proposed method coordinates all V2V links to improve the system-level performance. In the case of large-scale vehicular network, i.e., Fig. 8(c), the majority of the V2V links can satisfy the delay constraint under the proposed method; in other words, the transmission delay of the proposed algorithm does not exceed the threshold of 100 milliseconds. In contrast, the other two methods fail to meet the delay threshold of the V2V pairs.

G. Effectiveness of the DRL Training Performance

The D3QN model takes advantage of both Double DQN and the dueling architecture, enhancing the training performance and better evaluating the action value function. We compare the training loss of the V2V agents built by the D3QN and DDQN algorithms. As shown in Fig. 9, the D3QN model incurs lower estimation errors for the action value function, due to its better adaptation to the designed action space.

Fig. 10 compares the average cumulative reward of different DRL-based resource allocation methods, with the increasing number of training episodes. It is seen that the cumulative rewards are low and unstable at the beginning of the learning. With the increasing number of episodes, the average rewards grow, and then stabilize and converge. Moreover, with the increasing number of V2V agents, the number of iterations required to stabilize the cumulative reward also increases. It is observed that the proposed FedMARL method achieves faster convergence and higher rewards compared with the centralized SADRL mechanism.

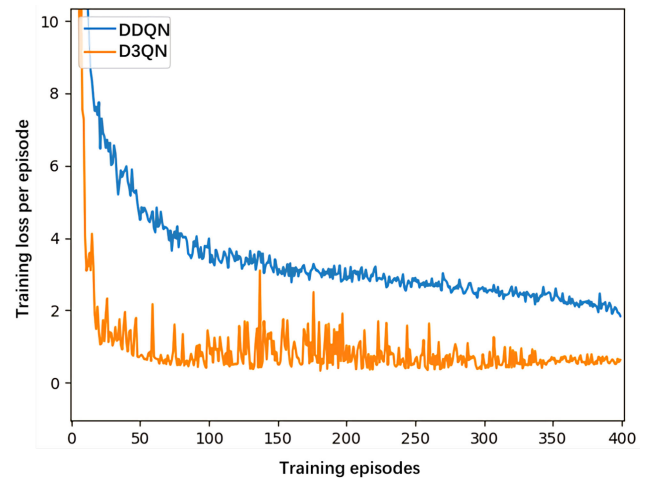


Fig. 9. Training loss of different DRL models.

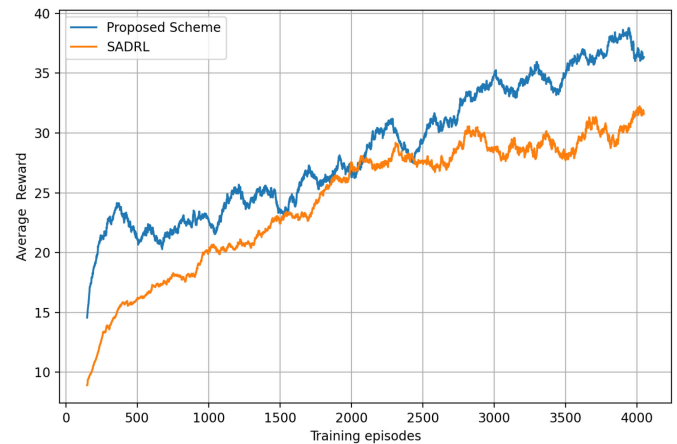


Fig. 10. The change of the average reward with the growing number of model training episodes.

VI. CONCLUSION

In this paper, we proposed a new FedMARL scheme to jointly optimize the channel selection and power allocation of V2V communication in a decentralized fashion. Specifically, the transmitter of each V2V pair acts as a learning agent and trains a local DRL model based on its local observation of channel conditions and interference. Different V2V agents synchronize their DRL models by using the FedAvg, thereby providing each individual agent with the global view of the network. Extensive simulations showed that FedMARL can satisfy the delay requirements of the V2V links while mitigating their interference to the cellular links.

In future works, we will investigate a more comprehensive system framework, where a V2V pair can switch adaptively between decentralized channel access (when the channel condition between the V2V pair is mild and no coordination of the BS is required, as designed in this paper) and centralized channel assignment (when the V2V link is obstructed or there are spare channel resources, and the BS can help coordinate the channel access).

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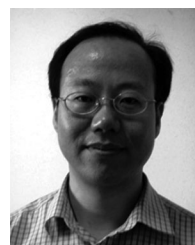


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